1. Handling Missing Data

DM 1ST.py - C:/Users/thala/Downloads/FODS/DM 1ST.py (3.14.3)

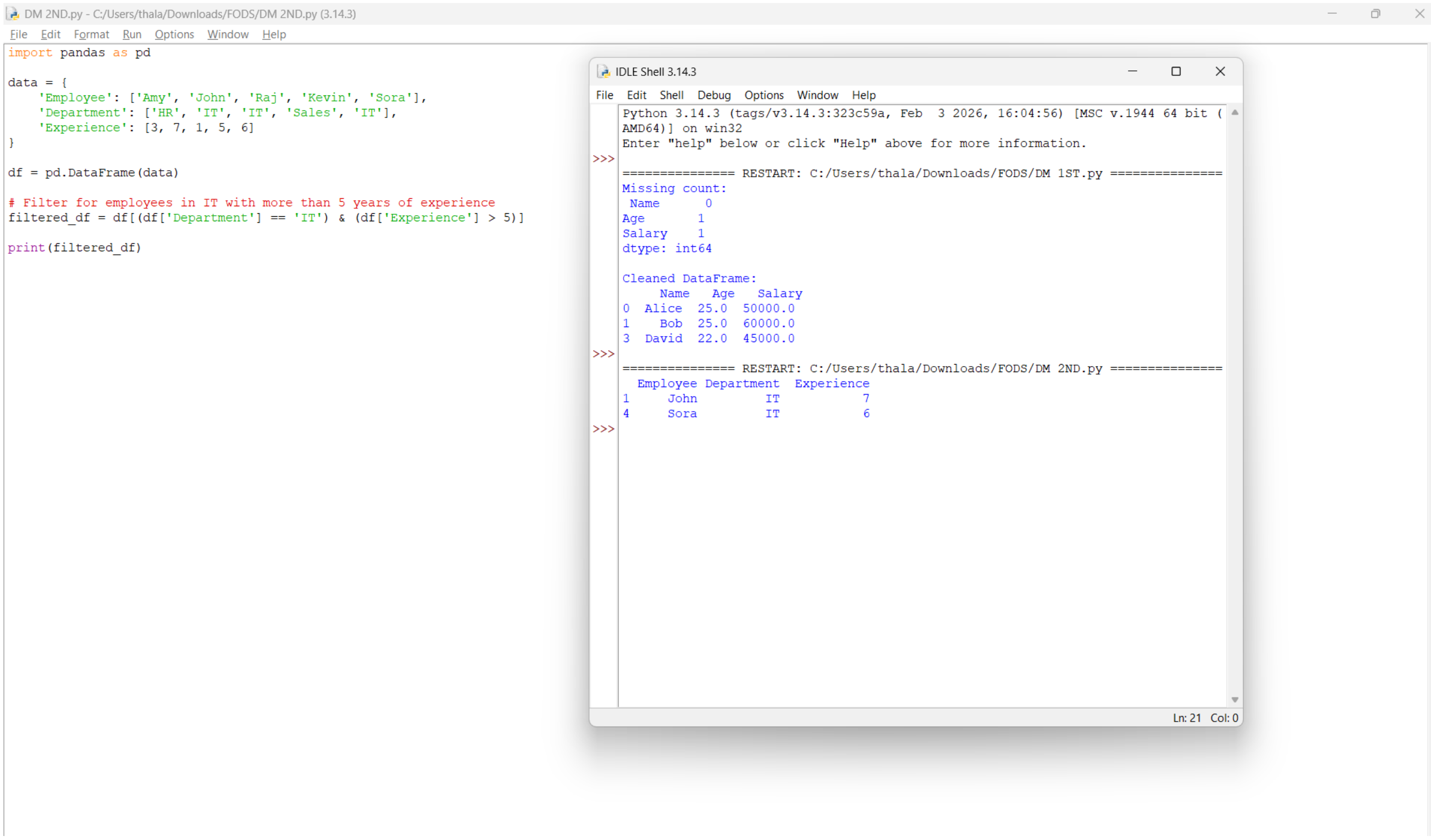
```python
import pandas as pd
import numpy as np

# Sample data with missing entries
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

df = pd.DataFrame(data)

# 1. Check for missing values
print("Missing count:\n", df.isnull().sum())

# 2. Fill missing Age with the column median
df['Age'] = df['Age'].fillna(df['Age'].median())

# 3. Drop rows where Salary is missing
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:\n", df)
```

IDLE Shell 3.14.3

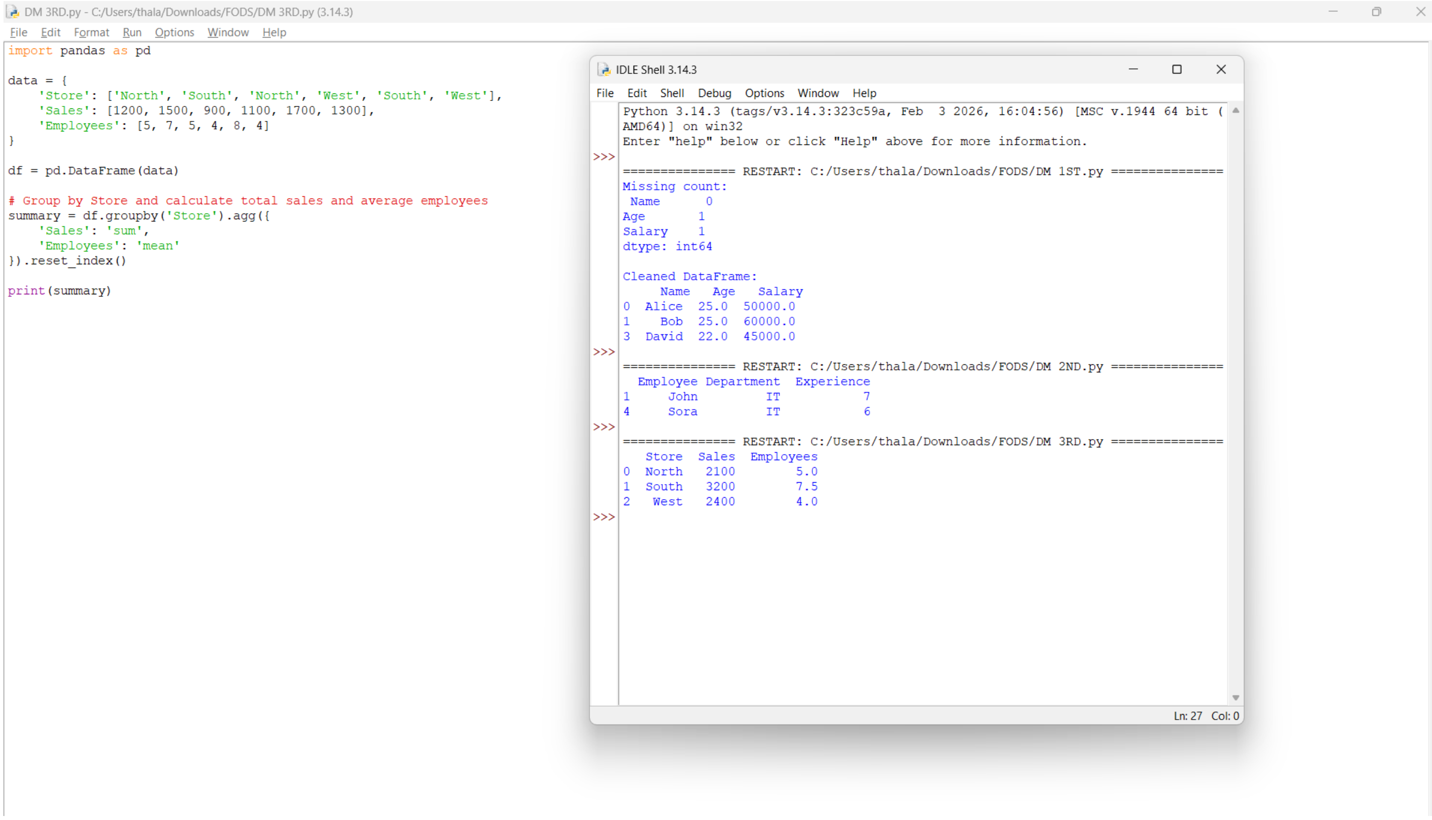
```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> 
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 1ST.py ===============
Missing count:
 Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
     Name   Age   Salary
0  Alice  25.0  50000.0
1    Bob  25.0  60000.0
3  David  22.0  45000.0
>>> 
```

Ln: 16 Col: 0

2. Filtering Rows Based on Multiple Conditions



3. Aggregating and Grouping Data



4. Python Program Using describe()

DM 4TH.py - C:/Users/thala/Downloads/FODS/DM 4TH.py (3.14.3)

```python
import pandas as pd

data = {
    'Age': [25, 30, 35, 40, 45],
    'Salary': [50000, 60000, 75000, 90000, 110000],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}

df = pd.DataFrame(data)

print("--- Numeric Summary ---")
print(df.describe())

print("\n--- Categorical Summary ---")
print(df.describe(include='str'))
```

IDLE Shell 3.14.3

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 4TH.py ===============
--- Numeric Summary ---
             Age         Salary
count   5.000000       5.000000
mean   35.000000   77000.000000
std     7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

--- Categorical Summary ---
       Department
count           5
unique          3
top            HR
freq            2
>>>
```

Ln: 22 Col: 0

5. Python Program Using head() and tail()